CGB Valuation Report

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1.Derive the portfolio value for the year of 2020 by using the data provided

1.1 Background analysis

Based on the provided data, this asset portfolio consists of six Chinese government bonds, all with fixed coupon rates. The bonds have varying issue dates (from 2016 to 2019), maturities (ranging from 2 to 10 years), and coupon rates (from 2.44% to 3.81%). The portfolio includes both short-term and medium-term bonds, with maturity dates spanning from 2021 to 2028. The majority of the bonds pay interest annually, except for two longer-term bonds (10-year maturities) that pay semi-annually. The portfolio's notional amounts show a mix of both long and short positions, with three bonds having negative principal values (-50, -20, and -30 million) suggesting potential short selling, and three bonds with positive principal values (5, 10, and 5 million). For now,we want to use their basic information and the specific information for yields from 2015 to 2020.

证券代码	证券简称	起息日期	债券期限(年)[单位]年	利率类型	票面利率(发行时)[单位]%	计息方式	计息基准	息票品种	到期日期	每年付息次数	本金金额（百万）
180001.IB	18附息国债01	2018-01-18	5.0000	固定利率	3.8100	单利	ACT/ACT	附息	2023-01-18	1	-50
180005.IB	18附息国债05	2018-03-08	7.0000	固定利率	3.7700	单利	ACT/ACT	附息	2025-03-08	1	-20
180019.IB	18附息国债19	2018-08-16	10.0000	固定利率	3.5400	单利	ACT/ACT	附息	2028-08-16	2	5
190003.IB	19附息国债03	2019-03-07	3.0000	固定利率	2.6900	单利	ACT/ACT	附息	2022-03-07	1	10
160023.IB	16附息国债23	2016-11-03	10.0000	固定利率	2.7000	单利	ACT/ACT	附息	2026-11-03	2	5
190002.IB	19附息国债02	2019-02-21	2.0000	固定利率	2.4400	单利	ACT/ACT	附息	2021-02-21	1	-30

1.2 Data Comprehension

This table represents a historical yield curve dataset for Chinese government bonds in 2020.The dataset lists various standard tenors, ranging from overnight to 50 years, along with their corresponding yields.

日期	标准期限说明	标准期限(年)	收益率(%) Yield to Maturity
2020-01-02	0d	0.00	1.2000
2020-01-02	1m	0.08	1.3442
2020-01-02	2m	0.17	1.8940
2020-01-02	3m	0.25	2.0092
2020-01-02	6m	0.50	2.1097
2020-01-02	9m	0.75	2.3235
2020-01-02	1y	1.00	2.4160
2020-01-02	10y	10.00	3.1485
2020-01-02	15y	15.00	3.4365
2020-01-02	2y	2.00	2.5316
2020-01-02	20y	20.00	3.4739
2020-01-02	3y	3.00	2.7458
2020-01-02	30y	30.00	3.7324
2020-01-02	40y	40.00	3.7680
2020-01-02	5y	5.00	2.9156
2020-01-02	50y	50.00	3.7837
2020-01-02	7y	7.00	3.0657
2020-01-03	0d	0.00	1.2000

Portfolio | 2020 yield curve | 2019 yield curve | 2018 yield curve | 2017 yield curve | 2016 y

The yield curves from 2016 to 2019 indicate that this 2020 yield curve is part of a larger historical collection of the portfolio.It would be used for valuation and risk management over time.

1.3 Introduction for Portfolio Valuation Methods

The value of a bond is the sum of the present value of all its future cash flows. These cash flows include:

(1)Regular payment of the interest.

(2)The principal repaid upon maturity.

The interest rate used to discount these cash flows is the treasury bond yield matching the remaining maturity of the bond.Having read the materials,we would like to use the Hermite model for matching yields respectively.

1.4 Specific Valuation

1.4.1 Data preprocess

(1) Tenor Standardization： Convert all tenor notations to decimal years;

eg: 12m -> 1y

(2) Year Fraction Computation： Count days to year as a fraction.

1.4.2 Hermite Linear Slope Calculation

Normally,we use Hermite to calculate the yields of bonds at some certain point.The cubic Hermite spline uses a cubic polynomial on each interval. Given two points at the interval $[t_0, t_1]$ with their function values P_0, P_1 and derivative values m_0, m_1, the interpolation polynomial $P(t)$ can be expressed as:

$$\begin{aligned} P(t) = {} & (2s^3 - 3s^2 + 1)P_0 \\ & +(s^3 - 2s^2 + s)(t_1 - t_0)m_0 \\ & +(-2s^3 + 3s^2)P_1 \\ & +(s^3 - s^2)(t_1 - t_0)m_1 \end{aligned}$$

where $s = \frac{t-t_0}{t_1-t_0}$ is the normalized parameter. This formulation ensures that:

$$P(t_0) = P_0, \quad P(t_1) = P_1,$$
$$P'(t_0) = m_0, \; P'(t_1) = m_1,$$

thus achieving exact matching of both function values and first derivatives.

1.4.3 Cash flow generation

The below funtion could calculate the cash flow at each payment point for the six bonds in the portfolio.

```python
def calculate_bond_cash_flows(self, bond_row, valuation_date):
    """计算单只债券的现金流"""
    valuation_date = pd.to_datetime(valuation_date)
    issue_date = bond_row['起息日期']
    maturity_date = bond_row['到期日期']
    coupon_rate = bond_row['票面利率(发行时/单位)'] / 100

    # 修复：先检查本金金额是否为NaN，再取绝对值
    face_value = abs(float(bond_row['本金金额（百万）'])) if not pd.isna(bond_row['本金金额（百万）']) else 0.0
    is_short = bond_row['本金金额（百万）'] < 0 if not pd.isna(bond_row['本金金额（百万）']) else False
    payment_freq = bond_row['每年付息次数'] if not pd.isna(bond_row['每年付息次数']) else 1

    # 检查日期有效性
    if pd.isna(issue_date) or pd.isna(maturity_date):
        return []

    # 计算剩余期限
    if valuation_date > maturity_date:
        return []  # 债券已到期

    cash_flows = []

    # 计算付息日
    current_date = issue_date
    payment_dates = []

    while current_date < maturity_date:
        if payment_freq == 1:  # 年付
            next_date = current_date + pd.DateOffset(years=1)
        elif payment_freq == 2:  # 半年付
            next_date = current_date + pd.DateOffset(months=6)
        else:  # 默认年付
            next_date = current_date + pd.DateOffset(years=1)
```

1.5 Conclusion

In conclusion,the portfolio value is -84.4 million.

```
1. 基本统计信息：
   观察期：2020-01-03 00:00:00 到 2020-12-31 00:00:00
   总交易日数：248
   当前组合价值：-84.40 百万

2. 10天99% VaR结果：
   相对VaR：-1.2096%
   绝对VaR：1.02 百万
   相对ES：  -3.5875%
   绝对ES：  3.03 百万

3. 收益率统计：
   平均日收益率：-0.0024%
   日收益率标准差：0.1745%
   最小日收益率：-2.1707%
   最大日收益率：0.5417%
```

2. Estimation of the 10-day 99% VaR for the above portfolio

2.1 Feasibility for historical analysis

The historical simulation approach is well-suited for this bond portfolio's VaR calculation,for the sufficient data from 2018 to 2020, which can directly capturing real interest rate dynamics and avoiding distributional assumptions.

Compared with Monte Carlo,it is claimed that the historical analysis uses less computational resources than Monte stimulation.

2.2 Excercise

From the aboved,we used historical stimulation for VaR calculation.The steps for 10-day 99% VaR is as followed:

(1)Obtain historical daily return data for all assets in the investment portfolio over a period of time.

(2)Based on these historical daily returns, the hypothetical 10-day returns of the investment portfolio over the past 500 days (assuming there are 500 days of data) are simulated and calculated. This is usually achieved by adding or compounding the daily returns of 10 consecutive days.

(3)Sort the 500 simulated 10-period returns from worst to best.

Since 1% of 500 samples corresponds to the fifth value, the loss amount corresponding to the fifth worst return after sorting is the VaR.

2.3 Conclusion

Year	Portfolio (M)	VaR (M)	VaR (%)	ES (M)
2020	-84.40	1.02	-1.21	3.03
2019	-84.96	0.43	-0.50	2.98
2018	-84.49	0.46	-0.54	0.51

3. Backtesting Conduction

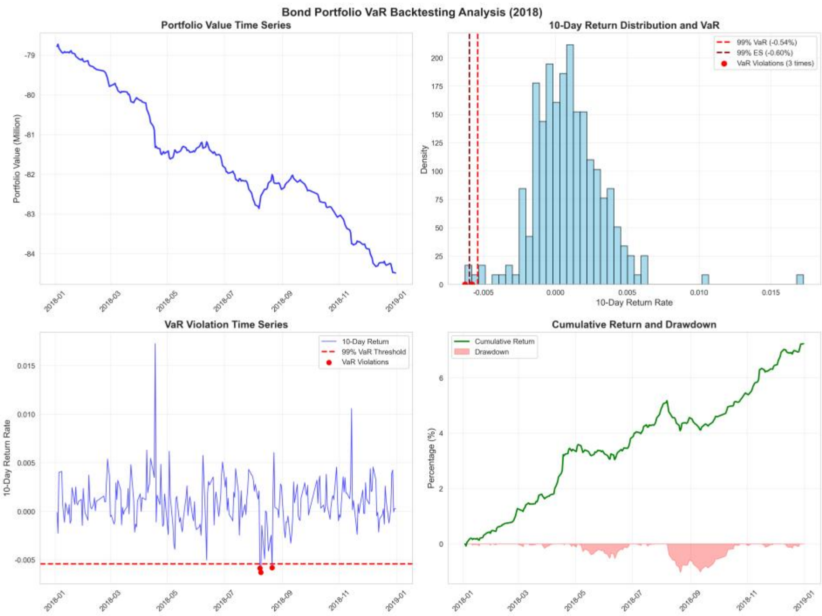
3.1 Results Interpretation

Year	Portfolio (M)	VaR (M)	VaR (%)	ES (M)	Nums of exception	Exception Ratio (%)
2020	-84.40	1.02	-1.21	3.03	3	1.21
2019	-84.96	0.43	-0.50	2.98	3	1.20
2018	-84.49	0.46	-0.54	0.51	3	1.20

From 2018 to 2020， the exception ratio is nearly about 1.2%.It indicates robustness of the risk.However,the VaR of 2020 increases over 0.7 pct,partly dued to the Covid-19.

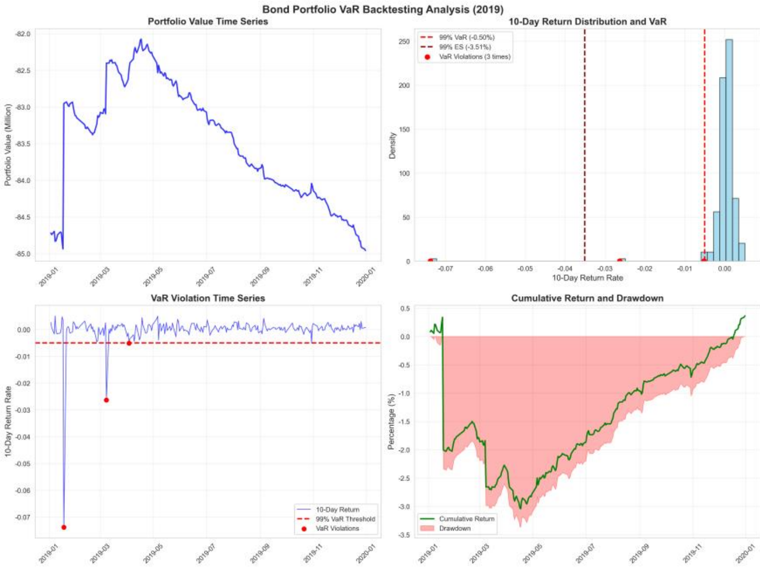
3.2 VaR for each year

3.2.1 2018 Value at Risk



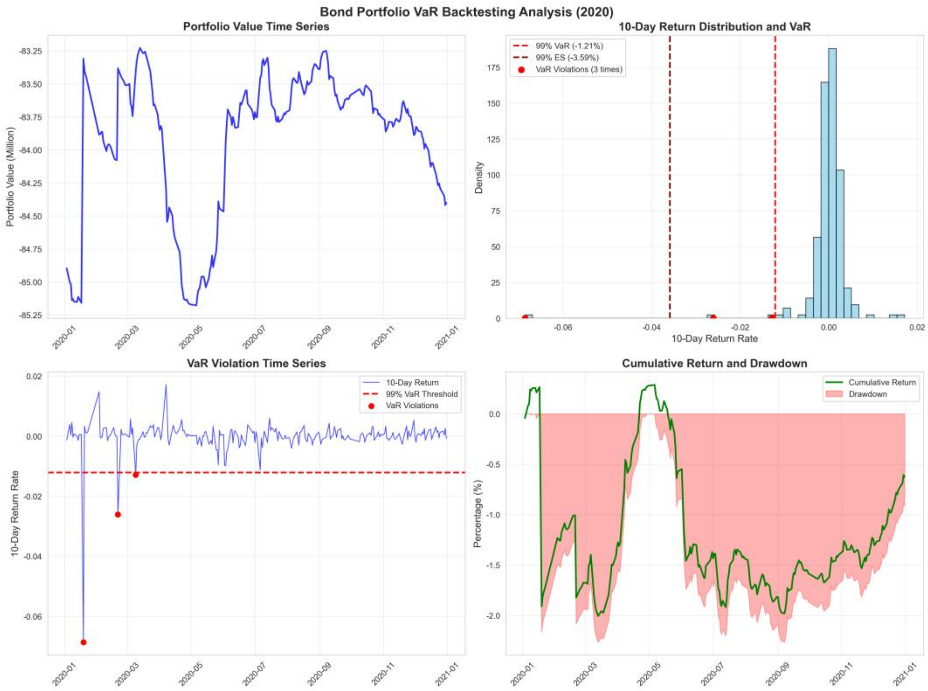
This chart presents a Value at Risk (VaR) backtesting analysis for the year 2018. The model uses a 99% confidence level to value the risk.The backtesting results show the distribution of 10-day portfolio returns, with the left tail representing losses.And we plotted 3 points as exceptions.The histogram also indicate that the frequency of returns is concentrated around zero, with a limited number of observations breaching the VaR threshold.By the observations of VaR Violation,we could know there are only 3 exceptions points,which reveals the efficiency of our risk management measures.

3.2.2 2019 Value at Risk



This chart displays the VaR backtesting results for 2019. The VaR is stated at +0.50%, and the Expected Shortfall (ES) at the 99% confidence level is -3.51%. The return distribution appears more spread out compared to the 2018 data, with a significant right tail indicating potential for higher gains, but also a left tail representing losses,suggesting that the model may have been more frequently challenged during this period, potentially due to higher market volatility.

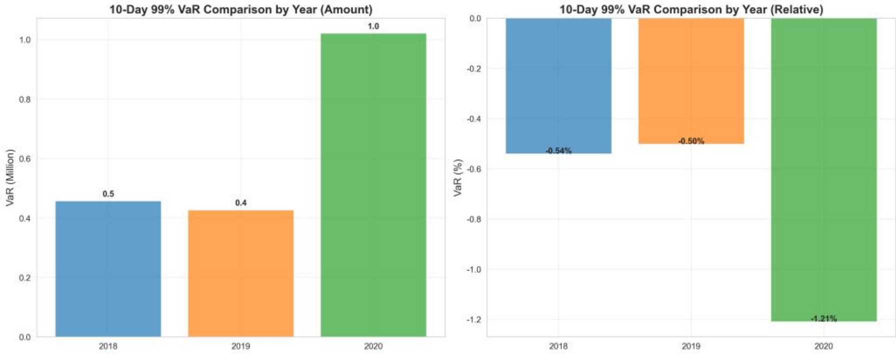
3.2.3 2020 Value at Risk



The backtesting results for 2020 highlight a year of higher volatility for the bond portfolio, with the VaR model experiencing several violations. This indicates that the model's risk estimates were occasionally exceeded by actual market movements, with a longer left tail in losses. The portfolio ended the year with a positive cumulative return, despite undergoing several drawdowns.

When February,Black Swan Events happened that the US financial market is facing multiple circuits with 2 apparent points in the graph 3. And it also indicates the high volatility of the return.

3.3 Improvements for futures



By comparison,we clearly observe that the risk of 2020 significantly increases.And we know Covid 19 erupted at the poin.Based on this,we have several advices for improvements of the models:

(1) Introduce Monte-Carlo Stimulation or other ML models as comparisons to get better performance;

(2) Consider the Black Swan event during building model

(3) Use more other perspective data for valuation.